Transfer: circuits and induction

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Apply circuit relationships and explain an induction observation.

SUCCESS CRITERIA - I can...

1. Use current and voltage rules in a new circuit.
2. Account for electrical power and energy.
3. Explain induction through changing flux.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “resistance” and “power” to explain your model or result. Sentence starters are optional.

resistance: voltage divided by current for a stated operating condition

power: rate of energy transfer

induction: producing an emf through a changing magnetic flux

01 Bring back an earlier idea

A motor is 62% efficient and the following generator is 71% efficient.
Find the combined useful fraction in sequence.

Word help: generator = a device converting mechanical input to electrical output

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $V = IR$; $P = IV$; $E = P\Delta t$; $|\text{emf}_{avg}| = N|\Delta\Phi|/\Delta t$. Use stated circuit or coil conditions. Ideal battery and wires unless stated otherwise.

Worked reasoning: A 12 V supply drives a 6 ohm resistor: current 2 A and power 24 W. A complete model checks both the charge-flow rate and the energy-transfer rate.

Your turn: Draw the circuit and an energy-flow arrow. **Label** what an ammeter and voltmeter would measure and where each connects.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Represent and interpret

Sketch two resistors in parallel across a battery and label the equal-voltage branches.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Circuit symbols: long/short lines = battery; rectangles = resistors.

Marzano check - circle: ① ② ③

04 Solve and justify

A 12 V battery powers 4 Ω in series with two parallel 8 Ω resistors.
Find total current, each branch current and total power.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Analyze the reasoning

A magnet is held motionless near a fixed coil, then moved into it.
Explain the change in the meter response without saying the magnet supplies unlimited energy.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

06 Choose a method

Measured (I, V) pairs are (0.10, 4), (0.20, 8), (0.30, 12) in A, V.
Find resistance and explain the evidence for an ohmic model.

Word help: ohmic = having constant resistance under the stated physical conditions

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

07 Exit check

A 50-turn coil has flux per turn change by .004 Wb in .20 s.
Find average emf magnitude.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A 12 V source feeds a 2 ohm series resistor and a parallel pair of 4 ohm resistors.
Find total power and power in each resistor; verify the energy-rate sum.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

Read down the left column, then down the right. Use the open space to draw, calculate and explain.